BENJAMIN PACHTER

Senior Data & AI Architect · Full-Stack AI Engineer

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Power systems AI engineer with hands-on production experience automating datacenter site selection and transmission capacity analysis at a major US utility. At Duke Energy, architected and delivered a geospatial capacity modeling platform responsible for 150+ hyperscaler data-center feasibility analyses, screening 100+ GW of large-load transmission requests across a 6-state territory — reducing load-study turnaround from 2–4 weeks to same-day and saving ~200 engineering hours/month. Builds production AI infrastructure end-to-end: GPU inference stacks (RTX 4090, CUDA, Ollama), autonomous agent systems (ReAct loops, multi-model routing, RAG), and full-stack geospatial platforms (Python/FastAPI + React/TypeScript). Uniquely positioned at the intersection of grid operations, geospatial ML, and applied AI engineering.

TECHNICAL SKILLS

LLMOps / GenAI	RAG, vector search (ChromaDB), prompt engineering, agent orchestration (LangChain), model serving, inference optimization, multi-model routing, fine-tuning (LoRA/QLoRA), Hugging Face, Ollama
ML / DS	PyTorch, TensorFlow, scikit-learn, XGBoost, Transformers/BERT, CNNs, time-series forecasting, feature engineering, model evaluation
Data / APIs	Python, SQL, Pandas, NumPy, SciPy, PySpark, Airflow, ETL/ELT, REST APIs, FastAPI, microservices, WebSockets; languages: Java, C++, C#, R, Scala, Bash, PowerShell; BI: Power BI (DAX/Power Query), Tableau
Cloud / Infra	AWS (S3, Lambda, Glue, Athena, Redshift, SageMaker, Bedrock), Docker, Kubernetes, CI/CD, Linux, CUDA, PostgreSQL, SQL Server, Railway, GitHub Pages
Geospatial	ArcGIS Pro, Python toolboxes, spatial raster modeling, interpolation (IDW), transmission capacity analysis, PSS/E integration, site screening
Frontend	React, TypeScript, Vite, Tailwind CSS, Zustand, Radix UI, Material UI (MUI), Emotion, Framer Motion, Three.js, Recharts, MapLibre GL
Domain Expertise	Power grid operations, transmission capacity analysis, grid interconnection queues, datacenter site screening, supply chain risk modeling, quantitative finance

EXPERIENCE

Duke Energy · Sep 2024 – Present
Senior Business Analyst · Charlotte, NC (Remote)

- Architected and delivered a spatial transmission-capacity modeling platform (ArcGIS Pro + Python toolbox) generating a continuous MW headroom raster (IDW interpolation on tower-level TrLim); screens 100+ GW of large-load requests across a 6-state territory
- Designing CIM-aligned, AI-ready data architecture and integration patterns to make grid/asset data interoperable and usable for downstream analytics, ML, and future LLM applications across the enterprise
- Developed prompt-driven and agentic workflows to automate site screening, feasibility narratives, and decision support; integrated structured enterprise data with LLM-based reasoning to reduce manual analysis loops
- Reduced PSS/E load-hosting study turnaround from 2–4 weeks to same-day by automating pre-screening and eliminating infeasible sites before engineering analysis; saved ~200 engineering hours/month
- Engineered reusable production data pipelines and services integrating AWS, SAP S/4HANA, Salesforce (Apex/Flows), and ArcGIS to support 5 internal organizations; automated operational reporting and a major industrial customer’s monthly billing pipeline

Architecture leadership: own platform roadmap, technical standards, and stakeholder alignment across transmission planning, economic development, and operations; translate ambiguous business needs into production-grade data/AI systems.

New York Mortgage Trust · Jan 2022 – May 2024
Acquisitions Analyst · Charlotte, NC

- Built multi-source data ingestion pipelines (Python, SQL) integrating Yardi and public-data APIs (FRED, FDIC, BLS, Census); delivered forecasting models (TensorFlow, XGBoost) for rent, occupancy, and cap-rate prediction to support institutional underwriting
- Deployed CNN-based document parsing and BERT/Transformer NLP pipelines for automated document understanding and sentiment analysis; migrated legacy workflows to cloud-native AWS/Azure services
- Delivered ML-driven risk scoring and executive analytics across a $7B+ residential mortgage portfolio; implemented Salesforce automations (Apex triggers, Flow) to streamline operations

Deloitte · Jan 2020 – Jan 2022
Risk & Financial Advisory Analyst · Seattle, WA

- Designed automated ETL pipelines (Python, AWS) and ML models (Random Forest, GBM, ARIMA, GARCH, LSTM) for portfolio risk forecasting and bond volatility prediction across finance, energy, and defense clients
- Built NLP pipelines for financial sentiment analysis across news and social media; supported SAP S/4HANA + Azure SQL migrations and delivered audit automation frameworks for Fortune 500 clients
- Developed stress-testing and Monte Carlo scenario models for fixed-income and commodity exposures; partnered with audit and risk teams to translate quantitative findings into client-ready deliverables across regulated finance, energy, and defense engagements

BMW Manufacturing · Jan 2018 – Dec 2018
Supply Chain Planning Co-op · Greer, SC

- Redesigned the JIT/JIS part-reorder flow — a dedicated low-volume milk-run for replacement parts freed every production truck to ship full — driving ~$13M in transportation cost reduction across the G05–G09 (X5 / X6 / X7 / XM) programs; built time-series forecasting models and dashboards (Python, Power BI) for supply-chain disruption risk

PORTFOLIO PROJECTS

Headroom · Interactive transmission-interconnection screen — B-θ DC power flow, PTDF sensitivities, and N-1 contingency sweep on a synthetic 14-bus grid; zero-dependency engine validated against hand-solved cases in an open test suite · live: bpachter.github.io/headroom · code: github.com/bpachter/headroom

Entropy · Scroll-driven interactive thermodynamics — seven physically accurate live sims (Carnot cycle, Maxwell's demon, Hawking evaporation) on a hand-written hard-sphere gas engine computing real quantities (S = k ln W, Landauer, Bekenstein–Hawking) over instanced WebGL · live: bpachter.github.io/entropy · code: github.com/bpachter/entropy

Gradient · Interactive journey into how machines learn — real in-browser backprop, tokenizer and attention visualizations, an n-gram language model with temperature / top-k / top-p sampling, LoRA / DPO / MoE explainers; zero dependencies · live: bpachter.github.io/gradient

Aether · GPU-native physics visualization museum; physically accurate CUDA C++ ray tracing of Schwarzschild and Kerr black hole spacetimes via direct geodesic integration — no game engine shortcuts; pre-baked WebM animations served via FastAPI/React, deployed on Railway

Thessa · Ontology-typed knowledge-graph intelligence engine — ~11,000 organizations and ~22K typed relationship edges over SEC filings and 78K+ federal contracts; autonomous extraction flywheel with independent citation-checking adjudication; hybrid local (RTX 4090) + cloud model fleet · product concept, in development

Duke GIS Capacity Heatmap Toolbox · Production ArcGIS Pro Python toolbox generating MW headroom raster (IDW interpolation on tower-level TrLim) for large-load site screening

EDUCATION

University of North Carolina at Charlotte
BSBA, Operations & Supply Chain Management and Management Information Systems · Cum Laude · December 2019

CERTIFICATIONS

IBM AI Engineering Professional Certificate · LLMs, Transformers, LoRA/QLoRA, RLHF, RAG, fine-tuning; PyTorch, TensorFlow, Hugging Face (13 courses)
DeepLearning.AI / Stanford — Machine Learning Specialization · Supervised/unsupervised learning, advanced algorithms, reinforcement learning
Microsoft Power BI PL-300 Data Analyst Associate · ETL, DAX, data modeling, Microsoft Fabric deployment
Architecting Solutions on AWS · Lambda, Glue, S3, Athena, Redshift, SageMaker
SAS Statistical Business Analyst · Regression modeling, statistical analysis, hypothesis testing